

QUOC-HUY NGUYEN

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EXPERIENCE

Data Scientist | GenAI | AWS | OpenAI | Utilities

Jun 2024 - Present

SoCalGas

- Leading the deployment, maintenance, and enhancement of a machine learning risk model that processes 1M+ records/year. Worked closely with engineering and operations to mitigate the 600+ gas pipeline damages per year, resulting in >\$1M savings while minimizing risks of injury or loss of life.
- Leveraged large language models (GPT-4o) and prompt engineering to structure and analyze 140k+ customer complaints, identifying actionable themes for product/service and systemic business process improvements. Built a multiprocessing pipeline accelerating complaint analysis by ~300%, enabling scalable insights generation.
- Migrated legacy corrosion analysis tool from Excel to Python and integrated AWS S3 data lake, reducing manual runtime by ~99% and enabling scalable, automated pipeline integrity assessments.
- Architected automated workflow to generate compliance letters for repeat offenders, cutting manual labor hours by ~98% and driving engagement from third-party companies to support pipeline and contractor safety goals.

Data Scientist (Freelancer) | Keras | Tensorflow | Consulting | Healthcare

May 2022 - Jun 2024

Upwork

- Finetuned a deep learning architecture (Keras EfficientNetV2) on noisy clinical EEG data, increasing seizure detection accuracy by 9% and enhancing patient outcomes.
- Diagnosed overfitting in a brain tumor medical imaging model (ResNet CNN) and advised regularization strategies, delivering interpretable and more generalizable models aligned with research requirements.
- Developed GeoPandas scripts to analyze malicious attack patterns, identifying ~10 high-risk geographic regions for targeted cybersecurity measures.

Machine Learning Researcher | LLMs | AWS | Streamlit | Youtube

May 2023 - Apr 2024

University of Michigan, School of Information

- [Developed](#) an end-to-end AI-powered product integrating Youtube's API, AWS infrastructure, an NLP/LLM backend (BERT, ChatGPT), and a user-facing Streamlit website for real-time viewing, to deliver personalized content recommendations.
- Configured a PostgreSQL database with user preference data, powering product telemetry and early-stage experimentation.
- [Explored](#) AI/ML in sports analytics to predict NHL draft rankings using unstructured scouting reports and word embeddings, demonstrating innovative NLP application to new problem domains.

Data Analyst | ETL | Fintech | Customer

Sep 2021 - Apr 2022

Curacao

- Engineered an ETL data pipeline to ingest 16+ GB of Open Banking data from FinTech APIs for personal loan risk assessment, streamlining underwriting for 1000+ of mobile app users.
- Analyzed key user segments from user transactions to optimize physical space utilization by ~5% across 14 department stores.
- Optimized SAS code logic for routine reports, reducing processing time by ~10%.

Undergraduate Researcher | Pytorch | Matlab | Single-cell Data Science

Jul 2019 - Mar 2021

Atwood Lab

- Implemented a Pytorch graph-based convolutional neural network (CNN) for semi-supervised classification model, based on a highly cited research article, to enhance cancerous cell identification accuracy. By leveraging innovative tools outside industry standards, such as principal graphs and graph neural networks, novel approaches were contributed to cancer research.

EDUCATION

Master of Applied Data Science, University of Michigan, Ann Arbor

- Awarded UMSI Millers Scholars Fund

Bachelor of Science in Mathematics, University of California, Irvine

- Data Science Specialization, Statistics Minor, Phi Beta Kappa, Summa Cum Laude

SKILLS

Programming: Python, SQL, PostgreSQL, R, SAS, C++, Matlab, Bash, Shell, Unix/Linux Terminal

Libraries: Numpy, Pandas, Polars, Scikit-learn, PyTorch, TensorFlow, Keras, PySpark, Statsmodels, NLTK, Matplotlib, Seaborn, Altair, Geopandas, Streamlit, ArcPy, Python-Docx, boto3

Datasets: Utilities, Customer, Financial, B2B, Healthcare/Clinical, Geospatial, Causal Inference, JSON, Experiments, Web Scraping

Tools: Git, DVC, AWS (S3, EC2, RDS, Sagemaker), Azure, Power BI, Domo, SAP HANA Studio, LaTeX, Agile Methodology